

CONDITIONAL NULLS AND POSITIVE CONTROLS FOR TASK-VECTOR SUBSPACE OVERLAP

Anonymous authors

Paper under double-blind review

ABSTRACT

Task-vector subspace overlap is often interpreted as shared learned structure because it exceeds an isotropic baseline, although linear-layer updates depend on the activations they read. We introduce a conditional randomisation that preserves each update’s spectrum and blockwise occupancy in frozen-base activation geometry while randomising cross-task directional agreement; its expected overlap has a closed form. We pair this null with a positive control that plants shared directions and measures recovery of the delivered signal, exposing consequential under-randomisation in our initial implementation. In complete ViT-B measurements, the corrected activation null reproduces 22–25% of above-isotropic overlap, with positive residual alignment across all fully fine-tuned pools. For shared-start LoRA, a regime-specific factor null reproduces about 80%, showing that the reference must match the training regime. The resulting decomposition is conditional rather than causal; the protocol makes structural claims auditable by requiring explicit conditioning assumptions and validation against planted alternatives.

1 INTRODUCTION

Independently fine-tuned models built on one pre-trained checkpoint have task vectors whose input-side singular subspaces overlap far above an isotropic random baseline, and a growing body of merging work measures that overlap and builds on it. [Gargiulo et al. \(2025\)](#) report the cross-task inner products of right singular vectors and define an interference score from them. [Zhou et al. \(2026\)](#) define a Right Subspace Overlap statistic over 190 pairs of task vectors and use it as one input to a predictor of merging success. [Qiu et al. \(2026\)](#) construct a null-space projector out of the cumulative update’s right singular vectors, and justify it with a theorem that task vectors approximately align with representation subspaces. In each case the overlap is compared, explicitly or implicitly, against what an unstructured subspace would give.

That comparison is correct for unconditional random subspaces, but it does not answer the conditional attribution question studied here. For a linear map $y = Wh$, the raw minibatch gradient has the outer-product form

$$\nabla_W L_t = \sum_{b \in B_t} g_{t,b} h_{t,b}^\top. \quad (1)$$

Under scalar-step gradient descent without weight decay, the final update’s row space therefore lies in the span of the activations along that specialist’s own training trajectory. This exact statement neither identifies a common frozen-base span nor applies unchanged after coordinatewise optimiser preconditioning: activations drift and optimisers transform gradients. It motivates a weaker, measurable question. Using the frozen base’s activation second moment as a common coordinate system, how much observed overlap is predicted after holding fixed each update’s own occupancy of that geometry? The isotropic value k/d_{in} conditions on none of it.

Prior work motivates both sides of this question. [Cheng et al. \(2025\)](#) relate linear-layer task vectors to their inputs, and [Qiu et al. \(2026\)](#) give an approximate subspace-alignment result; the latter note that the effect of shared initialisation remains insufficiently explored. These results do not establish frozen-base containment for the checkpoints studied here. Cross-task right-subspace overlap itself is already measured at scale by the work above. What is missing is a stated conditional baseline, an accounting of how much the answer moves under it, and evidence that the baseline randomises what it claims to.

We provide all three. The activation-conditioned null independently acts on each observed update with rotations that preserve its complete singular spectrum and blockwise read-mass profile in the frozen-base activation eigenbasis. Its expectation has a closed form: it is the overlap predicted by the two observed occupancy profiles. The remainder is residual directional agreement conditional on them. This is an operational conditional decomposition; the retained profile is measured after fine-tuning and may itself contain task-learned structure. A generative alternative sampled from C_H is not a valid conditional comparator because it must invent the update-activation relationship and produces a conspicuous level failure in our common-reference stress test.

Applied to the CLIP benchmarks used by this literature, the null reproduces 25.3% and 22.4% of above-isotropic top-8 overlap in the two complete ViT-B measurements. Conditional on the fixed calibration and blocking specification, the remaining excess stays positive throughout the crossed task-layer finite-pool stability interval. Under rank-16 LoRA, $\text{row}(BA) \subseteq \text{row}(A)$ makes the observed A factor a necessary regime-specific condition. Its top-8 right singular subspaces overlap at 0.88, and at 1.00 when linearisation keeps A bit-identical, while the learned B factor’s top-8 left singular subspaces overlap at 0.05 against 0.01 isotropic chance. An A - and factor-spectrum-conditioned null reproduces 80.3% of the product update’s above-isotropic overlap; although its attainable plant delivers only 37.4–57.0% of the nominal increment, the null recovers 100.7–101.1% of what is delivered. This one stress test does not identify which mechanism produced the conditioned structure.

The methodological result is why these estimates, rather than a larger number from our first implementation, are reported. A zero-signal level check accepts the identity map, so it cannot detect under-randomisation. We add a positive control that plants shared directions and measures the signal actually delivered and recovered. It exposed two under-randomisation defects in the activation null and a nuisance-spectrum mismatch in the LoRA null. Correcting the activation null raised delivered-signal recovery from 45.6–51.7% to 99.3–99.6% and moved the full- fine-tuning ViT-B/16 headline from roughly 67% to 25.3%. Surrogates independently drawn from that conditional null also form a relative stress test: the isotropic and participation-ratio-whitened baselines report +0.0171 and +0.0522 apparent excess. This comparison is conditional on our stated reference model and is not an independent intervention.

We make three contributions: a group-conditional decomposition of read-subspace overlap; a positive-control protocol that checks power as well as level and catches consequential defects in our own nulls; and a regime-specific audit of full fine-tuning and LoRA that separates effect-size attribution from pair ranking and downstream merge performance.

2 RELATED WORK

Subspace overlap in model merging. Merging grew from weight averaging (Wortsman et al., 2022; Ilharco et al., 2022; Matena & Raffel, 2022) through task arithmetic (Ilharco et al., 2023; Ortiz-Jimenez et al., 2023) to interference resolved coordinate by coordinate (Yadav et al., 2023; Yu et al., 2024) and then in subspaces (Stoica et al., 2025; Tang et al., 2023; Lee et al., 2025), which is where overlap became a reported quantity. Cheng et al. (2025) relate linear-layer task vectors to their corresponding inputs and Qiu et al. (2026) give an approximate subspace-alignment result. These works motivate activation-aware conditioning; they do not imply exact containment in a frozen-base subspace under feature drift and adaptive optimisation. The overlap measurement is established by Gargiulo et al. (2025), Zhou et al. (2026) and Marczak et al. (2025), none of which states a conditional reference distribution for it. Our contribution is that conditional baseline and a protocol for checking how much coupling it actually randomises.

Activation statistics as machinery. Activation second moments are long-standing, and continual learning has used them longest. Saha et al. (2021) build gradient subspace bases by SVD of stored input activations and Wang et al. (2021) project updates into the null space of the uncentered input feature covariance, the same object as our C_H . Both use it to constrain training. Inside merging the pattern repeats: RegMean (Jin et al., 2023) merges with per-model input Gram matrices computed on each fine-tuned model’s own data, and two data-free methods estimate the covariance from the parameter differences (Abdel Hameed et al., 2026; Xu et al., 2026). More directly, An et al. (2025) compute pre-linear-layer covariances from each task’s training examples and fine-tuned model to construct a knowledge-aware SVD for pruning task vectors. All target a task-conditioned object,

$\mathbb{E}_{x \sim \mathcal{D}_t}[xx^\top]$. We instead choose the frozen base under a recorded calibration mixture as a common, task-agnostic coordinate system; this is part of our estimand rather than a uniquely necessary null. Using $\Delta W^\top \Delta W$ itself as a reference would also be circular: exact inverse-square-root whitening equalises the nonzero singular values and makes a top- k choice non-identifiable rather than supplying an independent calibration geometry. What none supplies is a statement of what an uninformative task vector looks like, and the difference bites: RegMean’s published shrinkage returns $2.3\times$ chance on independent task vectors when repurposed as an evidential baseline (Appendix C).

Null models for similarity claims. Cui et al. (2022) show that representation-similarity metrics such as RSA and CKA (Kornblith et al., 2019; Raghu et al., 2017) are confounded by the population structure of the input data, giving “spuriously high similarity for even completely random neural networks”. Our conditional analysis is the weight-space analogue: there the shared object is population structure in the input distribution, while here the conditioned object is occupancy of a common activation geometry and the measured quantity is task-vector subspace overlap. Ding et al. (2021) make the complementary point for the measures themselves, arguing that a similarity metric must be judged on sensitivity to changes that matter and specificity against changes that do not, which is the pairing §4 imports for nulls. The general form is older. Elsayed & Cunningham (2017) formalise whether population-level structure exceeds what already-known primary features predict, by comparing against surrogate data that preserves those features and is otherwise maximally random. The null of §3 is that construction applied to weights, and §4 is the part of the surrogate-data tradition this literature has not imported: checking what the surrogate actually destroys.

Low-rank adapters. How a shared adapter basis should be read is what §5 bears on, and several methods construct or exploit one (Stoica et al., 2025; Panariello et al., 2025; Zeng et al., 2026). The geometric result has a functional counterpart already known. Zhu et al. (2024) show that in $\Delta W = BA$ the two factors are not interchangeable: fine-tuning B is substantially more effective than fine-tuning A , and a random untrained A performs nearly as well as a trained one. If A need not be trained to work, it is unsurprising that it carries little task-specific information. Ban & Ji (2026) reach the qualitative conclusion directly, reporting that the similarity between A matrices across adapters is attributable to the identical initialisation and not to shared knowledge. That conclusion is theirs. We add the geometric constraint, $\text{row}(\Delta W) \subseteq \text{row}(A)$, and a factor-conditioned null whose reproduced fraction we report with its measured response and scope.

3 A NULL FOR TASK-VECTOR SUBSPACE OVERLAP

3.1 WHAT IS BEING MEASURED, AND AGAINST WHAT

For a pair of task vectors $\Delta W_i, \Delta W_j \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ let $V_i^{(k)}$ hold the top- k right singular vectors of ΔW_i , and write

$$\mathcal{O}(i, j) = \frac{1}{k} \|V_i^{(k)\top} V_j^{(k)}\|_F^2, \quad (2)$$

the mean squared canonical correlation between the two read subspaces. For two independent uniformly random k -subspaces of $\mathbb{R}^{d_{\text{in}}}$ this has expectation k/d_{in} , which we verify numerically in Appendix B; that value is the isotropic null the literature reports against.

The question is not whether $\mathcal{O}$ exceeds k/d_{in} . It does, by an order of magnitude. We ask how much remains after conditioning on specified marginal structure in each update. Throughout, *null-reproduced* means reproduced after holding those stated statistics fixed. It does not identify which process created them: a statistic measured after fine-tuning can itself contain task-learned structure.

3.2 THE ACTIVATION-CONDITIONED NULL

Let $C_H = \mathbb{E}[hh^\top] = V \text{diag}(w) V^\top$ be the input second moment of the *frozen base model* at that layer. This is an operational reference geometry, not the claim that the final update is exactly contained in the frozen-base activation span: activations evolve during fine-tuning, and optimiser preconditioning can alter the elementary outer-product gradient geometry. Retain the first m_C eigenvectors and partition them into blocks $\{\mathcal{B}_b\}$, with orthogonal projectors P_b ; the unresolved complement is one additional block. Define

$$\mathcal{G} = \{Q \in O(d_{\text{in}}) : QP_b = P_bQ \text{ for every } b\}. \quad (3)$$

For each task independently, draw Q_i from Haar measure on this product of orthogonal groups and set $\widetilde{\Delta W}_i = \Delta W_i Q_i$. This preserves the full singular spectrum because $\widetilde{\Delta W}_i \widetilde{\Delta W}_i^\top = \Delta W_i \Delta W_i^\top$. If $\Pi_i = V_i^{(k)} V_i^{(k)\top}$ and $t_{ib} = \text{tr}(P_b \Pi_i)$, then, away from a singular-value tie at the rank- k boundary, it also preserves every block occupancy t_{ib} and every block energy $\|\Delta W_i P_b\|_F$. It randomises which directions within those allocations two tasks select. It also fixes richer blockwise Gram and principal-angle invariants listed in Appendix B, so the attribution is conditional on more than these summaries.

This construction makes the conditional estimand explicit. Independence and Haar invariance give

$$\mathbb{E}_{\mathcal{N}} \mathcal{O}(\widetilde{\Delta W}_i, \widetilde{\Delta W}_j) = \frac{1}{k} \sum_b \frac{t_{ib} t_{jb}}{\text{rank}(P_b)}. \quad (4)$$

Thus the scalar null-reproduced component is evaluated exactly from the two observed block-occupancy profiles, not estimated from random draws. For this activation null, sampled matrices are used only when actual null task vectors are required—for projector reconstruction, positive controls, and external-reference diagnostics—not for the scalar headline. For a reported pool, the null-reproduced fraction is $f_{\mathcal{N}} = (\overline{\mathcal{O}}_{\mathcal{N}} - \overline{k}/\overline{d_{\text{in}}})/(\overline{\mathcal{O}}_{\text{raw}} - \overline{k}/\overline{d_{\text{in}}})$; averaging over its stated rows precedes the ratio.

The blocks use a fixed operational coarsening. The exact commutant of empirical C_H rotates only its exact multiplicity eigenspaces. Our $\mathcal{G}$ is instead the commutant of a resolution-coarsened, truncated block operator: for a simple retained spectrum its retained action reduces to independent signs, while the deliberately unresolved complement is one full orthogonal block. Finite calibration data motivate merging nearby retained directions as well. We estimate eigenvalue errors from eight disjoint folds and form maximal consecutive runs connected by overlap of adjacent $\hat{\lambda}_j \pm 2\sigma_j$ intervals; Appendix C gives the exact split-sample formula. This is a fixed operational coarsening rule, not a confidence theorem or a unique null. At 1001 images, across all measured modules, both ViT-B encoders give a median 37 blocks over $m_C = 64$ directions; the complement is one full block. Appendix C reports a full $k/m_C/z$ sensitivity grid, the budget dependence, stored-spectrum limit, and rejected fixed-rank alternative; the rule, calibration sample, and measured power are all part of the specification.

Why not a generative null. A generative draw with input covariance proportional to C_H instead invents the update–activation relationship: it concentrates top- k mass toward high-variance activation directions and overlaps far more than the data in a common-reference diagnostic (Appendix N). Coordinate permutation fails for the opposite reason, destroying the occupancy profile despite passing the offset check.

3.3 THE NULL FOR LOW-RANK ADAPTERS

For a LoRA update $\Delta W = \frac{\alpha}{r} B A$ (Hu et al., 2022) the row space lies inside $\text{row}(A)$, and equals it whenever B has full column rank. At initialisation, adapters sharing A_0 therefore have the same admissible r -dimensional row space. This remains exact when A is frozen; when A is trained it need not, so we measure its final overlap rather than assume it. The activation null does not preserve this regime-specific constraint.

Our *factor-conditioned* null keeps each observed A and the singular values of B , and redraws $B \mapsto U \text{diag}(S_B) W^\top$ with independent Haar singular frames. It preserves A and B ’s spectrum, but generally not the product spectrum of BA ; the LoRA estimand therefore has different invariants from the activation null. The product-spectrum-preserving alternative over-randomises A ’s anisotropy, giving +0.2023 offset. Appendix I gives the algebra and alternatives. In the selected null’s positive control, the attainable row-space-constrained plant delivers 37.4–57.0% of its nominal increment and the null recovers 100.7–101.1% of that delivered increment. This establishes measured level and response for that intervention, but does not restore product-spectrum preservation.

3.4 ESTIMATING C_H

For the vision encoders studied here the exact $d_{\text{in}} \times d_{\text{in}}$ matrix is affordable and we use it. Appendix G gives a two-pass estimator for the sizes where it is not, and records why the calibration set is part of

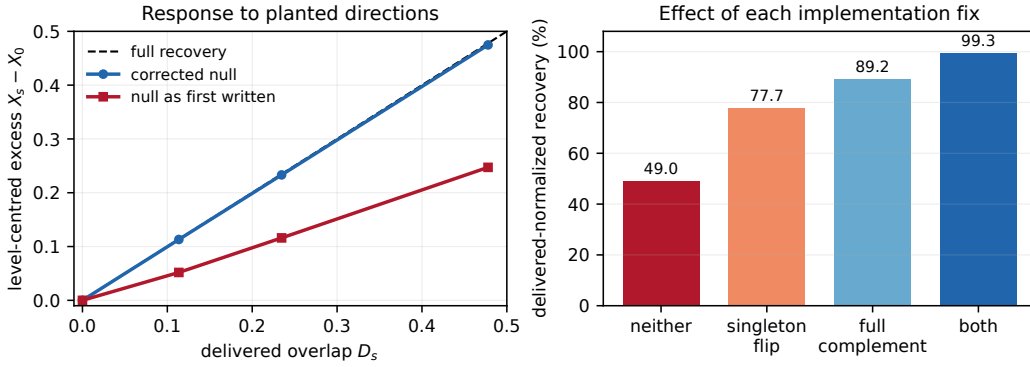


Figure 1: Left: level-centred excess reported against overlap actually delivered by the positive control. The diagonal is full recovery. Right: level-centred recovery at $s/k = 1/4$ with each activation-null correction enabled separately.

the measurement: C_H is an expectation over some input distribution, so we draw it from a mixture across several of the benchmark’s own datasets and record the composition with every covariance.

4 CHECKING A NULL

A null that changes nothing passes a zero-offset check while reproducing all measured overlap. This section pairs that necessary check with a positive control that can expose failure to randomise. We use “level” and “power” descriptively for null offset and planted-signal recovery, not for a rejection threshold, test size, or rejection probability.

4.1 LEVEL IS NOT ENOUGH

The natural check on a null passes a null that does nothing. Build a matched reference pair from an explicitly specified null generator, pass it through the candidate randomisation, and confirm that the reported excess has zero offset. Call this the H_0 check. We report external finite-draw diagnostics separately from the exact scalar decompositions.

Write (A_0, B_0) for a pair from the constructed reference and let $\mathcal{N}_1, \mathcal{N}_2$ be independent candidate-null draws. The H_0 statistic is

$$\mathbb{E}[\mathcal{O}(A_0, B_0) - \mathcal{O}(\mathcal{N}_1(A_0), \mathcal{N}_2(B_0))], \quad (5)$$

and for $\mathcal{N} = \text{id}$ this is exactly zero. A Haar-orbit self-reference is likewise idempotent in distribution, so its population H_0 is exactly zero. The recorded external H_0 instead uses independently activation-generated, spectrum-matched endpoints (Appendix M); because that generator is not the conditional orbit law, its offset mixes reference mismatch, over-destruction, and Monte Carlo error. It is only a stress diagnostic and is never added to the estimand. The identity null can therefore earn a perfect score while randomising little of its target. The asymmetry is not incidental. Eq. 5 measures over-destruction, the error that makes a null report structure where there is none. Under-destruction makes a null report that real coupling as null-reproduced. The check is blind to that failure mode. It is also relative to the constructed reference; it neither establishes a unique data-generating null nor supplies a model-free false-positive rate. The missing diagnostic is a check that can fail under an alternative.

4.2 POWER, MEASURED BY PLANTING

The complement of the H_0 check is a positive control. Take a real task vector A and a partner P_0 drawn from the fitted null, then replace s of the partner’s top- k read directions with directions from A , projection/QR-complete the remaining directions, and retain P_0 ’s singular values to obtain P_s (exact operator in Appendix L). The intervention introduces forced directional matches on top of the pair’s pre-existing overlap; finite-dimensional cross terms mean that it need not add exactly s/k .

Table 1: Native excess on the same independent-orientation reference pairs, ViT-B/16, $k = 8$, eight fixed modules, and 20 pairs per module (160 total; seed 0). Exact orthogonality subtracts zero; isotropic subtracts k/d ; whitened rows use the participation-ratio floor or all 64 stored directions before subtracting k/d ; the final three rows subtract their own surrogate mean. Thus entries share overlap units but not an estimand. Our row is circular and is shown only as an implementation check.

baseline	reported excess
exact orthogonality (Gargiulo et al., 2025)	+0.0275
isotropic k/d	+0.0171
whitened at the participation ratio	+0.0522
whitened, all stored directions	+0.0554
generative activation model	−0.8959
coordinate permutation (paired action)	−0.0000
activation-conditioned (ours, circular here)	−0.0007

We therefore normalise by what the intervention actually delivers. For candidate null $\mathcal{N}$ define

$$D_s = \mathbb{E}[\mathcal{O}(A, P_s) - \mathcal{O}(A, P_0)],$$

$$X_s^{\mathcal{N}} = \mathbb{E}[\mathcal{O}(A, P_s) - \mathcal{O}(\mathcal{N}_1(A), \mathcal{N}_2(P_s))], \quad R_s^{\mathcal{N}} = \frac{X_s^{\mathcal{N}} - X_0^{\mathcal{N}}}{D_s}. \quad (6)$$

Recovery is thus level-centred and uses delivered rather than nominal overlap. We run this semi-synthetic stress test on real task vectors and the measured calibration covariance. The plant preserves singular values but can change the block-occupancy profile on which the null conditions, so R_s measures sensitivity to this intervention after reconditioning; it is not power against every possible cross-task dependence or evidence about how the conditioned statistics arose.

Figure 1 separates the two arms. The null as first written recovers only 45.6–51.7% of delivered overlap while returning +0.00004 at $s = 0$, which looks well behaved under the offset check; the corrected null recovers 99.3–99.6% and has −0.000623 at $s = 0$. Both arms score the same planted pairs and differ only in the randomisation.

4.3 WHAT THE POSITIVE CONTROL FOUND

The diagnostic exposed two under-randomisation failures in the activation null: its fixed-rank complement rotation reached only 4–18% of that space, and it skipped singleton sign flips although those directions carry 71% of stored eigenvalue mass. The LoRA generator instead matched only $\|B\|_F$, letting an uncontrolled factor spectrum set even the sign of its offset. The zero-offset check missed the first two and had been declared rather than measured for the third. Appendices I and K give the fixes and ablation.

4.4 A COMMON-REFERENCE STRESS TEST FOR EXISTING BASELINES

To expose how existing baselines behave on the same reference pairs, we form semi-synthetic pairs by independently applying our activation-conditioned generator to real updates. Residual orientation coupling is absent by construction under this reference, while each endpoint retains its observed spectrum and occupancy profile. Table 1 reports each baseline’s native excess; these are diagnostics on a common input distribution, not one shared estimand, because whitening changes the matrices being scored. Because our generator defines the reference, this experiment neither independently validates our null nor establishes model-free false-positive rates.

The untransformed reference pair’s raw overlap is 0.0275, or 2.6 times isotropic chance; subtracting the isotropic comparator reports +0.0171 excess. Whitening changes the scored matrices and reports still larger native excesses (Table 1). A transform introduced to reduce interference inside a merge therefore need not be calibrated as an evidential baseline.

The coordinate permutation is the useful counterexample. Here the same permutation is applied to both endpoints, so overlap, and hence its offset, is exactly unchanged; this paired action is a deliberate counterexample rather than the independent action in Eq. 5. It nevertheless destroys occupancy

relative to fixed C_H , so §3.2 rejects it. Passing an offset check is not sufficient, which is §4.1’s lesson from the other side.

4.5 THE SAME TEST ON THE LORA NULL

For LoRA, we score only the attainable plant $P_s^* = P_s A^+ A = (\alpha/r) B_s A$, with $B_s = (r/\alpha) P_s A^+$; both raw and null terms use P_s^* (Appendix E). It delivers 37.386%, 49.790%, and 57.043% of the nominal affine increment at $s = 1, 2, 4$. Normalising the level-centred excess by those delivered increments gives 101.117%, 101.078%, and 100.701% recovery. The initial norm-matched generator also has near-100% level-centred response, but its unplanted level is -0.1485 , versus -0.0073 after factor-spectrum matching. The correction therefore calibrates level rather than rescuing low power or an upward bias. This remains one intervention over four modules and six tasks.

4.6 WHAT THE CORRECTION COSTS

For the activation null, delivered-signal recovery rises from 45.6–51.7% to 99.3–99.6% while the unplanted offset moves from $+0.00004$ to -0.000623 . This is the desired combination: small reference offset and high response to the planted alternative. The study covers four modules and one family of interventions, and planting can alter conditioned occupancies, so we do not transfer the sub-percent deviation as a correction or bound on the 72-module observational estimate.

5 EXPERIMENTS

Setup. We evaluate CLIP vision encoders fine-tuned on the task-arithmetic benchmark of Ilharco et al. (2023) as extended to twenty tasks by Wang et al. (2024): all 190 pairs over all 72 linear modules for each ViT-B encoder, and 48 q/v modules for ViT-L/14. C_H comes from 1001 images drawn round-robin across seven of the benchmark’s own datasets on the frozen base; a separate rebuilding record stores source counts, fold membership, and hashes. All substantive overlap results use $k = 8$. For whitening comparisons we truncate at the module’s participation ratio m_{white} ; the activation null retains $m_C = 64$ eigendirections and reads its rotation blocks from the eigenvalue standard errors as in §3. Appendix C reports how the stored dimension and block structure move with the calibration budget. At fixed source counts, two source-stratified ViT-B/16 covariance refits give 25.102% and 25.302% versus 25.345%; the maximum 0.243-percentage-point shift is an observed two-refit diagnostic, not a calibration confidence interval. A deliberately different balanced MNIST/Cars/SVHN calibration gives 21.482% (-3.863 points); its residual stays positive under the same finite-pool resampling. Thus the residual’s sign is robust in this control, but the reproduced fraction is composition-sensitive. The external-reference H_0 is a separate stress diagnostic and is never added to an estimate. Brackets are 95% finite-pool stability intervals: the 2.5th and 97.5th percentiles of 50,000 crossed task-layer resampling replicates (seed 20260915). They describe sensitivity to reweighting the observed tasks and layers, not superpopulation coverage, and condition on one checkpoint/training realisation per task, the realised rows, fixed calibration and blocking rule, and the exact scalar null (or the pooled 64-draw realised Monte Carlo null for adapter arms; Appendix A).

5.1 FULL FINE-TUNING: THE CONDITIONAL NULL REPRODUCES ABOUT A QUARTER

In the two complete ViT-B measurements, the activation-conditioned null reproduces 25.3451% ([19.3033, 31.8010]) and 22.3961% ([15.5516, 29.8850]) of above-isotropic top-8 overlap. The partial ViT-L/14 q/v measurement gives 11.8961% ([8.1064, 16.4895]). Conditional on the fixed null specification, the residual is positive under crossed task-layer resampling, but should not automatically be identified with task-derived sharing: it is directional agreement not accounted for by the conditioned statistics. An earlier implementation put its headline reproduced fraction near 67%; the positive control in §4 shows why we reject it, with delivered-signal recovery of only 45.6–51.7% instead of the corrected null’s 99.3–99.6%.

Figure 2 shows the split; Appendix M gives the aggregate columns behind it. In the exploratory ViT-B/16 grouping, the exact null reproduces 16.5568% for modules reading the residual stream and 38.9609% for modules reading a representation built within the block. This contrast is not

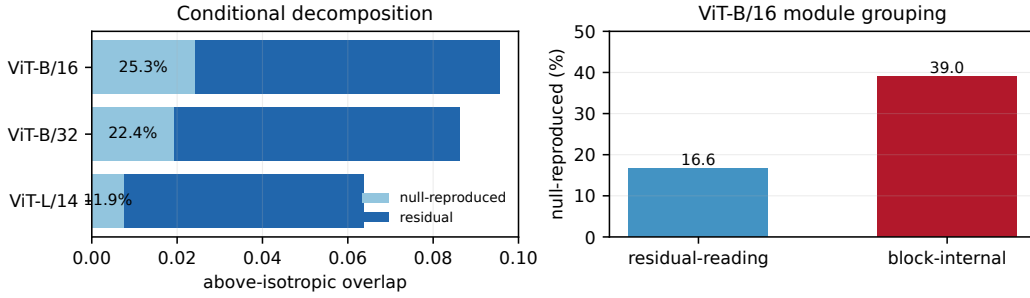


Figure 2: Left: above-chance overlap split into the part the activation null reproduces and the residual for each encoder. Right: an exploratory ViT-B/16 split by whether a module reads the residual stream or a representation built within its block. All fractions use the null expectation without adding the realised H_0 diagnostic.

explained by input width alone because attention $q/k/v$ and `out_proj` have the same input width. One possible explanation is that task-specific structure accumulates in the residual stream, whereas `out_proj` and `fc2` read representations more tightly constrained by the block. This is a descriptive grouping; it does not identify why the split arose.

For ViT-B/16, ViT-B/32, and ViT-L/14 respectively, the crossed-resampling residual stability intervals are $[+0.04540, +0.10568]$, $[+0.04224, +0.10085]$, and $[+0.03476, +0.08252]$. All endpoints remain positive under this finite-pool reweighting. Within each replicate, independent multinomial endpoint counts C_t and layer counts L_ℓ give row weight $C_a C_b L_\ell$; all module types in a layer stay together, weighted means precede the ratio, and external H_0 diagnostics remain separate.

5.2 LOW-RANK ADAPTERS: A SHARED-START-COMPATIBLE SIGNATURE

Low-rank adapters make the regime-specific conditioned object explicit. Since $\text{row}(BA) \subseteq \text{row}(A)$, shared A_0 imposes a common admissible row space; this remains exact when A is frozen, while for trained A we measure rather than assume it. Across all 24 targeted modules in the factor audit (Appendix D), learned-from-zero B overlaps at 0.0540–0.0570, well below A ’s 0.8759–1.0000 but still about five times chance. The linearized pool, with bit-identical frozen A , has the highest A and product overlap. The following pool-level estimates use all 24 targeted modules and eight tasks. For LoRA-16, the A -conditioned factor null reproduces 80.3110% ($[75.7205, 85.0681]$) of above-isotropic overlap. This conditions jointly on observed final A and B ’s spectrum. The attainable plant delivers 37.39/49.79/57.04% of its nominal increment, while level-centred recovery is 101.12/101.08/100.70% at $s = 1/2/4$. This establishes response to that narrow intervention. With frozen bit-identical A , the linearized pool gives 76.0713% ($[72.1974, 80.2483]$). Full fine-tuning over the same 24 modules gives 16.2440% ($[9.2333, 27.4812]$), leaving a 64-point contrast with the module set held fixed.

A regime-mismatched null changes the attribution. The activation null instead leaves 99.9% of the same LoRA pool’s above-chance overlap as residual, versus 19.7% under the observed- A -preserving factor null ($+0.5966$ versus $+0.1176$). It answers a different question because it does not hold observed A fixed; transferring a null across regimes therefore changes the inferred effect size substantially.

An independently initialised- A language-model study instead finds shared directions in B (Tang & Yang, 2026); its different architecture, tasks, and protocol preclude a direct comparison.

5.3 CONSEQUENCES: ATTRIBUTION RATHER THAN A NEW MERGE

We rebuild the null-space projector of Qiu et al. (2026) from real and independently randomised task vectors (Appendix J). Observed-null basis overlap is 0.8750 versus 0.0208 chance for LoRA and 0.2594 versus 0.1667 for full fine-tuning. The corresponding basis mass in the top activation subspace is 0.1356 and 0.2612 (chance 0.0833); none of these projector statistics has a dedicated positive

control. This does not test projector utility. A separate cross-term-only compression heuristic loses to the cumulative-update basis at every rank (Appendix F). Calibration is also nearly rank-preserving over task pairs (Spearman 0.994–0.995); we do not test a retrained mergeability predictor. The contribution is therefore inferential: a large effect or useful basis does not by itself establish the mechanism invoked to explain it.

6 DISCUSSION

Cross-task top-8 sharing remains positive after conditioning in every fully fine-tuned CLIP pool, but its attribution changes. The null reproduces 25.3% and 22.4% in the complete ViT-B runs; the factor-conditioned LoRA value is 80.3%, with near-full recovery of delivered signal in its narrow positive control. These quantities split agreement under stated conditions; they do not locate its origin in the base or the shared start.

A reader should report a null’s invariants, training regime, reference offset, and positive-control response. Here conditioning changes effect size while pair rankings barely move, and a separate coupling-only basis does not improve merge accuracy. A structural claim is only as specific as the null it survives and only as credible as that null’s measured response.

REPRODUCIBILITY STATEMENT

Appendix H records task pools, calibration data, numerical settings, and checks; the supplementary artifact provides row-level results, executable analysis, and manifests.

USE OF LARGE LANGUAGE MODELS

Large language models were used for methodology and experiment feedback, code implementation, mathematical and numerical checking, result interpretation, figure and artifact preparation, and drafting and editing. The authors reviewed and tested the resulting work and take full responsibility for the paper’s content.

REFERENCES

- Marawan Gamal Abdel Hameed, Derek Tam, Pascal Jr Tikeng Notsawo, Colin Raffel, and Guillaume Rabusseau. Model merging via data-free covariance estimation. *arXiv preprint arXiv:2604.01329*, 2026.
- Bang An, Yibo Yang, Philip Torr, and Bernard Ghanem. Purifying task vectors in knowledge-aware subspace for model merging. *arXiv preprint arXiv:2510.14697*, 2025. doi: 10.48550/arXiv.2510.14697.
- Hao Ban and Kaiyi Ji. Rethinking parameter sharing for LLM fine-tuning with multiple LoRAs. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. arXiv:2509.25414.
- Runxi Cheng, Feng Xiong, Yongxian Wei, Wanyun Zhu, and Chun Yuan. Whoever started the interference should end it: Guiding data-free model merging via task vectors. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 10121–10143, 2025. URL <https://proceedings.mlr.press/v267/cheng25h.html>.
- Tianyu Cui, Yogesh Kumar, Pekka Marttinen, and Samuel Kaski. Deconfounded representation similarity for comparison of neural networks. In *Advances in Neural Information Processing Systems*, volume 35, 2022. arXiv:2202.00095.
- Frances Ding, Jean-Stanislas Denain, and Jacob Steinhardt. Grounding representation similarity with statistical testing. In *Advances in Neural Information Processing Systems*, 2021.
- Gamaleldin F. Elsayed and John P. Cunningham. Structure in neural population recordings: an expected byproduct of simpler phenomena? *Nature Neuroscience*, 20(9):1310–1318, 2017. doi: 10.1038/nn.4617.

- Antonio Andrea Gargiulo, Donato Crisostomi, Maria Sofia Bucarelli, Simone Scardapane, Fabrizio Silvestri, and Emanuele Rodolà. Task singular vectors: Reducing task interference in model merging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Gabriel Ilharco, Mitchell Wortsman, Samir Yitzhak Gadre, Shuran Song, Hannaneh Hajishirzi, Simon Kornblith, Ali Farhadi, and Ludwig Schmidt. Patching open-vocabulary models by interpolating weights. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.
- Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. Dataless knowledge fusion by merging weights of language models. In *International Conference on Learning Representations*, 2023.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97, pp. 3519–3529, 2019.
- Yu-Ang Lee, Ching-Yun Ko, Tejaswini Pedapati, I-Hsin Chung, Mi-Yen Yeh, and Pin-Yu Chen. STAR: Spectral truncation and rescale for model merging. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*, 2025.
- Daniel Marczak, Simone Magistri, Sebastian Cygert, Bartłomiej Twardowski, Andrew D. Bagdanov, and Joost van de Weijer. No task left behind: Isotropic model merging with common and task-specific subspaces. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 43177–43199, 2025. URL <https://proceedings.mlr.press/v267/marczak25a.html>.
- Michael S Matena and Colin A Raffel. Merging models with fisher-weighted averaging. In *Advances in Neural Information Processing Systems*, volume 35, pp. 17703–17716, 2022.
- Guillermo Ortiz-Jimenez, Alessandro Favero, and Pascal Frossard. Task arithmetic in the tangent space: Improved editing of pre-trained models. In *Advances in Neural Information Processing Systems*, 2023.
- Aniello Panariello, Daniel Marczak, Simone Magistri, Angelo Porrello, Bartłomiej Twardowski, Andrew D. Bagdanov, Simone Calderara, and Joost van de Weijer. Accurate and efficient low-rank model merging in core space. In *Advances in Neural Information Processing Systems*, 2025.
- Zihuan Qiu, Lei Wang, Yang Cao, Runtong Zhang, Bing Su, Yi Xu, Fanman Meng, Linfeng Xu, Qingbo Wu, and Hongliang Li. Null-space filtering for data-free continual model merging: Preserving stability, promoting plasticity. In *International Conference on Learning Representations*, 2026.
- Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, pp. 6076–6085, 2017.
- Gobinda Saha, Isha Garg, and Kaushik Roy. Gradient projection memory for continual learning. In *International Conference on Learning Representations*, 2021.
- George Stoica, Pratik Ramesh, Boglarka Ecsedi, Leshem Choshen, and Judy Hoffman. Model merging with SVD to tie the Knots. In *International Conference on Learning Representations*, 2025.

- Anke Tang, Xianglin Luo, Li Shen, Yong Luo, Liang Ding, Han Hu, Bo Du, and Dacheng Tao. Concrete subspace learning based interference elimination for multi-task model fusion. *arXiv preprint arXiv:2312.06173*, 2023.
- Anke Tang, Li Shen, Yong Luo, Yibing Zhan, Han Hu, Bo Du, Yixin Chen, and Dacheng Tao. Parameter efficient multi-task model fusion with partial linearization. In *International Conference on Learning Representations*, 2024.
- Yixuan Tang and Yi Yang. Crowded in B-space: Calibrating shared directions for LoRA merging. *arXiv preprint arXiv:2604.16826*, 2026.
- Ke Wang, Nikolaos Dimitriadis, Guillermo Ortiz-Jimenez, François Fleuret, and Pascal Frossard. Localizing task information for improved model merging and compression. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Shipeng Wang, Xiaorong Li, Jian Sun, and Zongben Xu. Training networks in null space of feature covariance for continual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 184–193, 2021.
- Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162, pp. 23965–23998, 2022.
- Bo Xu, Haotian Wu, Hehai Lin, Weiquan Huang, Beier Zhu, Yao Shu, and Chengwei Qin. ACE-merging: Data-free model merging with adaptive covariance estimation. *arXiv preprint arXiv:2603.02945*, 2026.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. TIES-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, 2023.
- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Siqi Zeng, Yifei He, Meitong Liu, Weiqiu You, Yifan Hao, Yao-Hung Hubert Tsai, Makoto Yamada, and Han Zhao. Task vector bases: A unified and scalable framework for compressed task arithmetic. *Transactions on Machine Learning Research*, 2026.
- Luca Zhou, Bo Zhao, Rose Yu, and Emanuele Rodolà. Demystifying mergeability: Interpretable properties to predict model merging success. *arXiv preprint arXiv:2601.22285*, 2026.
- Jiacheng Zhu, Kristjan Greenewald, Kimia Nadjahi, Hartz Sáez de Ocariz Borde, Rickard Brühl Gabrielsson, Leshem Choshen, Marzyeh Ghassemi, Mikhail Yurochkin, and Justin Solomon. Asymmetry in low-rank adapters of foundation models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 2024. URL <https://proceedings.mlr.press/v235/zhu24c.html>.

A LIMITATIONS AND FUTURE WORK

All quantitative conclusions concern CLIP vision encoders. The headline uses $k = 8$, $m_C = 64$, and the 2σ resolution convention; we sweep k , m_C , and the resolution multiplier below. ViT-L/14 covers only $\mathbb{Q}$ and $\mathbb{V}$, and C_H is estimated from seven of the twenty benchmark datasets under one recorded mixture. One deliberately different composition is tested below; broader calibration compositions and language models remain open. The reported finite-pool stability intervals hold the calibration draw, fold partition, block rule, and m_C fixed; they do not incorporate uncertainty from estimating or selecting the null. They are not superpopulation coverage statements: each task contributes one checkpoint/training realisation, so training-run and checkpoint variation are unmeasured. Scalar activation-null values use Eq. 4 exactly; adapter intervals hold their pooled 64-draw realised Monte

Carlo null values fixed. We draw 50,000 crossed task–layer resampling replicates. Independent multinomial endpoint counts C_t and layer counts L_ℓ give each row weight $C_a C_b L_\ell$; all module types within a selected layer remain grouped. Each replicate takes the ratio after weighted averaging. The pseudorandom seed is 20260915. As a finite-pool influence check for the exact activation arms, leave-one-task-out reproduced fractions range 24.685–25.896%, 21.856–22.939%, and 11.310–12.165% for ViT-B/16, ViT-B/32, and ViT-L/14; the corresponding leave-one-layer-out ranges are 24.359–28.411%, 21.272–26.392%, and 10.955–13.587%. Every residual remains positive.

The raw outer-product gradient identity involves activations along each fine-tuning trajectory and, under scalar-step gradient descent, constrains the update to their span. Our null instead conditions on the frozen base’s second moment. Feature drift, adaptive coordinatewise preconditioning, and weight decay break an exact passage between those statements. Consequently, the activation-conditioned decomposition is descriptive: its post-fine-tuning occupancy profile can include learned structure, and identifying its origin would require an intervention or an additional stability assumption that we do not test.

The LoRA result is limited to rank 16, one shared-start protocol, eight tasks, and the reported attention modules. Its factor null preserves A and the singular spectrum of B , not the product spectrum of BA . In a four-module, six-task stress test the attainable plant delivers 37.4–57.0% of its nominal increment, and the null recovers 100.7–101.1% of that delivered signal. This establishes level and response only for that planting operator; the 87.5% projector-basis fraction has no dedicated positive control. A controlled shared-versus-independent- A_0 experiment is absent, and B remains about five times above isotropic chance. Finally, calibration is nearly rank-preserving and the cross-term basis does not improve compressed-merge accuracy; the evidence supports measurement and attribution claims, not a better merging algorithm.

B INVARIANTS OF THE OVERLAP STATISTIC

Every numerical property below is checked directly in the analysis; these checks should accompany the released measurement code rather than be inferred from library defaults.

Beyond the scalar occupancies and energies in §3.2, the group action fixes $\Delta W_i P_b \Delta W_i^\top$ and the eigenvalues of $V_i^{(k)\top} P_b V_i^{(k)}$ for every block. The former contains output-side block Gram geometry and the latter its principal-angle spectrum. A looser null conditioning only on t_{ib} could therefore give a different attribution.

Why Eq. 4 is exact. Haar averaging within block b gives $\mathbb{E}[Q_i^\top \Pi_i Q_i] = \sum_b (t_{ib}/d_b) P_b$, where $d_b = \text{rank}(P_b)$. For independent task-wise draws, linearity and $P_b P_c = \mathbf{1}\{b = c\} P_b$ give $\mathbb{E} \text{tr}(Q_i^\top \Pi_i Q_i Q_j^\top \Pi_j Q_j) = \sum_b t_{ib} t_{jb} / d_b$; division by k yields Eq. 4.

- **Orientation.** `top_right` returns the right singular vectors, of shape (d_{in}, k) , matching an explicit SVD on a rectangular matrix where a transpose would be silent. The convention $\Delta W = U S V^\top$ puts V on the input side; swapping it reverses every conclusion in this paper.
- **Normalisation.** $\mathcal{O}(A, A) = 1$ to 10^{-6} , and two independent uniformly random k -subspaces average 0.03995 against $k/d = 0.04000$ over 300 draws. We confirm the same identity at $(d, k) \in \{(200, 8), (200, 32), (400, 8), (3584, 8), (100, 10)\}$.
- **Degenerate inputs are refused.** The SVD of a zero matrix returns an arbitrary orthonormal basis, and returns the *same* one each time, so two zero task vectors score $\mathcal{O} = 1.0000$. In a LoRA checkpoint every module the adapter does not target is exactly zero, so an unguarded run over an adapter pool fills its averages with the maximum possible value taken from modules that did not change. `top_right` raises on such an input and the caller drops and counts those cells.
- **Checkpoint reads.** Weights are read by matching module paths to checkpoint keys by suffix, since `CLIPVisionModel.named_modules()` yields `encoder.layers.0...` while the checkpoint stores `vision_model.encoder.layers.0...`. A suffix matching more than one key is refused: the full CLIP checkpoint holds both `text_model.*` and `vision_model.*` under identical suffixes, and picking either silently would build task

vectors from the text tower and compare them against a vision covariance. Tensors returned this way are bit-identical to a `state_dict` load.

C CALIBRATION CHOICES FOR WHITENING AND ACTIVATION BLOCKS

Block-error rule. Partition the calibration set into eight disjoint folds and compute the ordered covariance eigenvalues $\lambda_j^{(f)}$ independently on each fold. With their mean $\bar{\lambda}_j^F$, the implementation uses the split-sample standard error $\sigma_j = [\sum_{f=1}^8 (\lambda_j^{(f)} - \bar{\lambda}_j^F)^2 / (8 \cdot 7)]^{1/2}$. The blocks are the transitive consecutive components induced by overlap of adjacent $\hat{\lambda}_j \pm 2\sigma_j$ intervals, where $\hat{\lambda}_j$ is the eigenvalue from the full calibration pool. This is an operational coarsening, not a confidence interval for eigenspace identifiability.

Finite calibration refits. Holding the seven source counts and extraction procedure fixed, two source-stratified bootstrap refits (seeds 20260916 and 20260917) give full ViT-B/16 exact reproduced fractions 25.102% and 25.302%, versus the 25.345% headline, for a maximum 0.243-percentage-point shift. This is the observed range of two refits, not a calibration confidence interval, and does not test a different source composition.

Composition stress control. Separately, we sample without replacement from deterministic test splits to balance 334 MNIST, 334 Stanford Cars, and 333 SVHN images (seed 20260919). This deliberately different calibration gives exact ViT-B/16 null 0.029621, residual 0.074953, and reproduced fraction 21.482%, or 3.863 percentage points below the seven-source headline. Holding this covariance fixed, the crossed finite-pool stability intervals are [16.8975, 26.2690]% for the fraction and [+0.048606, +0.109015] for the residual. The median participation ratio is 12.691, with 71/72 modules below 64, so both composition and spectral effective rank have changed. This is a stress control, not an estimate of calibration uncertainty or an interval over possible compositions.

A complementary post-hoc stratification asks whether zero, one, or both endpoints of a task pair belong to the seven calibration sources (78/91/21 pairs). The reproduced fractions rise from 23.06/26.56/28.33% for ViT-B/16, 20.38/23.45/24.55% for ViT-B/32, and 10.20/12.64/14.47% for ViT-L/14; the residual remains positive in every cell. This gradient is consistent with composition sensitivity, but it does not isolate calibration membership from task identity and is not used for inference.

Truncation. Whitening two task vectors by a shared C_H manufactures overlap when it is aggressive. On independent task vectors whose true overlap is chance (0.0400), a Tikhonov ridge at $\lambda = 10^{-4}\bar{w}$ returns 0.7199 and RegMean’s published off-diagonal shrinkage $\alpha = 0.9$ returns 0.0931. The mechanism is that $C_H^{-1/2}$ amplifies small eigendirections and drives both task vectors onto the same few of them: at $\lambda = 10^{-4}\bar{w}$, 96.5% of the whitened read mass lands on the twenty smallest stored eigendirections. Whitening only the top m_{white} directions keeps reference overlap near chance (0.0394, 0.0405, 0.0422 at $m_{\text{white}} = 8, 32, 64$).

We set m_{white} per module from that module’s participation ratio $(\sum_i w_i)^2 / \sum_i w_i^2$, used here as an operational effective-rank truncation. In our calibration refits, directions beyond this cutoff were unstable; the participation ratio is not an identifiability threshold. The ratio must be computed from the full spectrum: taken over only the m_C retained eigenvalues it is bounded above by m_C by Cauchy–Schwarz, which understates it and makes any gate comparing the two a tautology.

Why a block count was rejected. §3 sets the rotation blocks from what the calibration data can resolve. We record here what the alternative looked like, because the alternative is what a reader would reach for first.

Fixing a block count and complement rotation rank leaves two choices that no invariant selects, and the answer moves with them: sweeping the rotation rank of the complement from 32 to 512 moved the reproduced fraction from 48% to 23% under the null as it then stood. Subtracting that null’s offset on a constructed reference did not stabilise it. The resolution-based construction eliminates those two outcome-sensitive choices, since the complement is a single unresolved block and is rotated in full. It

does not eliminate design choices: the uncertainty multiplier, fold-SE scheme, calibration sample, and stored dimension remain part of the null specification.

The sweep is also where the defect of §4.3 hid in plain sight. A rotation rank that changes the answer by a factor of two is a rotation rank that is destroying different amounts of structure, and the endpoint the construction actually calls for, rotating the whole complement, lies outside the range that was swept.

Block structure is a function of the calibration budget. Blocks are the groups of eigendirections that our operational interval rule treats as unresolved. They tend to sharpen as the calibration set grows: estimated eigenvalue errors contract, so fewer adjacent intervals overlap and blocks split. This behavior does not make interval overlap a confidence theorem for eigenspace identifiability. The first three columns below are measured on nested subsets of one image pool. At each budget we partition images into eight disjoint folds and compute the eight foldwise eigenvalue replicates used by the split-sample rule in §3. The remainder scales the measured standard errors, the pool having stopped at 1001.

calibration images	measured			extrapolated		
	224	500	1001	2000	5000	∞
ViT-B/16	16	25	37	46	55	64
ViT-B/32	15	21	37	44	55	64

This is a resource-resolution trade-off that the method makes visible. Too few images and the blocks are coarse, the null rotates across gaps that are in fact resolved, and its constructed-reference offset grows. As the budget grows, the rule tends to split blocks. Under a simple population spectrum and fixed $m_C = 64$, its retained-region limit is $\{\pm 1\}^{64}$ while the unresolved complement remains $O(d_{\text{in}} - 64)$. Thus the limit is not the identity: signs randomise off-diagonal terms within the retained region, and the complement is fully rotated. It has no power against agreement expressed only as shared retained-axis occupancy, while retaining power against sign-sensitive or complement orientation alignment. The ∞ column extrapolates the retained block count only; it is not a measured power result.

The budget matters for a reason that is visible in the block counts themselves. A run at 224 images drawn independently of the pool above resolved 13 and 11 blocks where the nested 224 subset resolves 16 and 15, so at that budget the block structure is not stable across draws of the calibration set, let alone across architectures. At 1001 the two architectures happen to agree at a median of 37 blocks and the same median largest block. This is a useful cross-architecture check, not proof of sampling stability. The calibration budget is part of the measurement specification and must be fixed and reported.

The reproduced fraction is not reported against the budget, since the two budgets were measured under nulls that differ by more than the budget: the 224-image run predates the corrections of §4.3, so comparing them would charge the calibration set for a change that belongs to the null.

Where the blocked region ends. The activation-null stored dimension m_C divides the space the null treats two ways: inside $\text{span}(V)$ it rotates within resolvable blocks, outside it rotates in full. We sweep $k \in \{4, 8, 16, 32\}$, $m_C \in \{16, 32, 64\}$, and uncertainty multiplier $z \in \{1, 2, 3\}$ on each reported pool:

pool	$k = 8$, all (m_C, z)	full (k, m_C, z) grid
ViT-B/16	20.22–25.61%	13.28–30.37%
ViT-B/32	17.94–22.55%	13.31–28.51%
ViT-L/14 q/v	5.84–12.00%	5.36–19.01%

At fixed (k, m_C) , the largest observed change from varying z is 0.754 percentage points; most of the fixed- k range comes from m_C . This is variation in the conditional estimand—storing more directions changes what the null holds fixed—rather than numerical instability of its exact estimator.

We pre-specify the headline at $m_C = 64, z = 2$; m_C remains distinct from the per-module whitening truncation m_{white} above.

The calibration must use the observed spectrum. The reference offset is not a property of the null alone; it depends strongly on the spectrum of the task vectors the null is calibrated against. Substituting a synthetic exponential spectrum for the observed one raises the realised reference offset at eight blocks from $+0.0041$ to $+0.0254$, a factor of six. This sensitivity is another reason not to transfer that diagnostic into the estimand. The reason is that real task vectors have rapidly decaying spectra, which makes the top- k subspace well determined and the block rotation effective, whereas a flat spectrum leaves that subspace ill-determined and the block structure dominates. A null calibrated against a spectrum the data does not have calibrates the wrong quantity.

D LORA FACTOR AUDIT

Table 2: ViT-B/16, eight tasks, 28 pairs, all 24 targeted attention modules; all entries use top-8 singular subspaces. B is initialised at zero and is therefore entirely learned; A is not. `lin-LoRA` is the partial linearization of [Tang et al. \(2024\)](#), which leaves A bit-identical across tasks.

	A right	B left	ΔW right
LoRA-16	0.8759	0.0540	0.6075
lin-LoRA	1.0000	0.0570	0.6649
isotropic	0.0104	0.0104	0.0104

At the $k = 8$ boundary, the relative singular gap $(s_8 - s_9)/s_8$ is above 10^{-3} for every observed A and BA cell in both eight-task, 24-module adapter pools. Across 24,576 factor-null BA redraws, only four fall below 10^{-3} and none below 10^{-4} ; the median gap is about 0.099 in each arm. Thus exact or numerical cutoff ties do not drive the aggregate factor-null result, although the finite- k choice remains part of the estimand.

E POWER OF THE LORA NULL

Table 3: A rank-16 LoRA stress test, ViT-B/16, four attention modules, six tasks. The fitted factor null has pre-planting overlap $\mathcal{O}_0 = 0.4825$. From an unconstrained plant P_s we form the attainable $P_s^* = P_s A^+ A = (\alpha/r) B_s A$, where $B_s = (r/\alpha) P_s A^+$. Both raw and null terms score P_s^* . Delivery divides Δ_{raw} by $(s/k)(1 - \mathcal{O}_0)$; recovery is level-centred as in Eq. 6.

	planted	raw	delivery	initial X_s	corrected X_s	recovery
0/8	0.4825	—	—	-0.1485	-0.0073	—
1/8	0.5067	37.386%	—	-0.1243	+0.0172	101.117%
2/8	0.5469	49.790%	—	-0.0840	+0.0579	101.078%
4/8	0.6301	57.043%	—	-0.0009	+0.1414	100.701%

The initial norm-matched generator also has approximately 100% level-centred response, but a large $s = 0$ error. Spectrum matching changes that level from -0.1485 to -0.0073 while retaining near-full response. The correction therefore fixes level; it is not evidence that the initial null had low power or upward bias. Values just above 100% reflect finite draws.

F MERGED ACCURACY BY RANK BUDGET

The separate cross-term-only heuristic uses $\sum_{t \neq s} \Delta W_t^\top \Delta W_s = \tau^\top \tau - \sum_t \Delta W_t^\top \Delta W_t$. This does not improve compressed-merge accuracy: the τ basis wins at every tested budget, the gap widens with r , and at $r = 256$ it wins on all eight tasks separately. Concentrating empirical cross-task coupling is therefore insufficient for better compressed-merge accuracy in this experiment.

Table 4: Merged accuracy averaged over the eight tasks, ViT-B/32, 500 test images per task, update compressed to rank r . Zero-shot is 38.9% and uncompressed task arithmetic is 59.5%.

rank r	8	16	32	64	128	256
τ basis	45.9	49.9	52.8	55.7	57.6	59.0
cross-term basis	45.3	48.8	51.4	53.5	54.8	55.7

Calibration is close to rank-preserving. Across the 190 ViT-B/32 pairs, raw and calibrated overlap have Spearman 0.995 and share 17 of the top 20 pairs; ViT-B/16 gives 0.994 and shares 19 of the top 20. Thus an ordering-only consumer receives nearly the same ordering on these pools, although we do not retrain the mergeability predictor of Zhou et al. (2026).

G ESTIMATING C_H

C_H is $d_{\text{in}} \times d_{\text{in}}$, which is 1.4 GB per module at $d_{\text{in}} = 18944$ and infeasible to store across a network. For the vision encoders studied here the exact matrix is affordable (594 MB across all 72 modules of ViT-B/32) and we use it. For larger models we provide a two-pass estimator: a Frequent Directions sketch locates the subspace, and a second pass accumulates the exact $m_C \times m_C$ second moment inside it. The second pass is necessary. Frequent Directions biases eigenvalues downward by construction, and whitening applies $C_H^{-1/2}$, so an understated eigenvalue is an over-amplified direction; on synthetic data at sketch size 128 the eigenvalue error falls from 0.49 to 0.0016 between the two passes while the top-8 subspace agreement is 0.998 either way.

The calibration set is part of the measurement. C_H is an expectation over some input distribution, and the answer depends on which. We draw it from a mixture across several of the benchmark’s own datasets, since a single one would make C_H the base geometry *on that task’s inputs* and reintroduce the conditioning the test exists to remove. A separate rebuilding record lists all image identifiers, per-source counts, hashes, and fold memberships, while each covariance manifest records its item count and a basename–byte-size hash. We verified that the two records enumerate the same 1001 basenames; the release links them explicitly rather than implying that source labels are embedded in each flattened covariance manifest. The covariances themselves come to 26 GB and need not be distributed; a reproducibility artifact must include this rebuilding record and the extraction commands.

H REPRODUCIBILITY RECORD

Checkpoints and task pools. The frozen bases are openai/clip-vit-base-patch16, openai/clip-vit-base-patch32, and openai/clip-vit-large-patch14. Full-fine-tuned specialists follow tanganke/<architecture>.<task>. The 20 task labels are cifar10, cifar100, dtd, emnist_letters, eurosat, fashion_mnist, fer2013, food101, gtsrb, kmnist, mnist, oxford-iiit-pet, oxford_flowers102, pcam, rendered_sst2, resisc45, stanford-cars, stl10, sun397, and svhn. The matched eight-task comparison uses dtd, eurosat, gtsrb, mnist, resisc45, stanford-cars, sun397, and svhn; its adapter repositories append _lora-16 or _l1-lora-16. Rank and scale are read from each adapter_config.json, and untargeted modules are dropped rather than treated as zero subspaces.

Calibration and numerical settings. The rebuilding record contains 143 images from each of cifar10, dtd, eurosat, gtsrb, resisc45, stl10, and sun397; fold sizes are 126, 125, 125, 125, 125, 125, 125, 125. Images are the first valid examples encountered in the streaming training split of each tanganke/<source> dataset, converted to RGB. Extraction uses each model’s CLIPImageProcessor, all sequence tokens (class and patch), exact uncentred $\sum hh^T/n$ accumulation in float32, $m_C = 64$, and CPU execution. The extractor-input basename/size hash common to the three covariance manifests has prefix 8f88002f7557; the separate logical image/fold record has identifier-list-hash prefix 73797f525986. Full digests are in the artifact.

Randomisation and verification. The activation-null scalar results are the exact expectation in Eq. 4; they use no random draws. On the matched full-fine-tuning pool, a sampled check gives null 0.02330 versus exact 0.02333. External-reference diagnostics use finite draws, the two factor-null adapter arms use four independent 16-draw batches (seeds 0, 1000003, 2000003, and 3000017), pooled row-wise to 64 draws, and projector rebuilds use six. The batch-based pooled Monte Carlo standard errors (sample SD divided by $\sqrt{4}$, three degrees of freedom) are 0.0067 and 0.0054 percentage points for LoRA-16 and linearized LoRA-16; pooling changes their seed-0 16-draw fractions by -0.0109 and -0.0093 points. The common-reference baseline uses 20 pairs for each of eight fixed modules at seed 0. `analysis/check_paper_numbers.py` recomputes point estimates from row-level CSVs, and `analysis/overlap_bootstrap.py` reproduces the 50,000 crossed-resampling stability intervals. The artifact pins immutable revisions for all bases, specialists, adapters, and calibration datasets, together with a locked runtime environment. The original B/16 covariance manifest omitted its base-revision label; its two candidate revisions are tensor-identical in all 400 tensors, with content hashes recorded, so the numerical weights are unambiguous. Large model weights, covariance tensors, and calibration images are not bundled.

I HOW THE THREE DEFECTS WERE FIXED

The complement was rotated inside a fixed frame. ΔW splits into a part inside $\text{span}(V)$ and a residual R whose rows lie in the complement. The first version drew a random 128-dimensional frame of that complement and rotated R inside it, leaving the remainder bit-identical between two task vectors. The complement is $d_{\text{in}} - m_C$ dimensional, so this covered 18% of it at $d_{\text{in}} = 768$ and 4% at $d_{\text{in}} = 3072$, while about three quarters of the real read mass lives there. The fix needs no square Haar matrix: R has rank at most $\min(d_{\text{out}}, d_{\text{in}} - m_C)$, so a Haar map on the complement carries its right singular vectors to a uniform frame of that rank, which one thin SVD and one QR deliver exactly.

Singleton blocks were skipped. Blocks of size one were left alone on the grounds that a one-dimensional space cannot be rotated. Its orthogonal group is $\{+1, -1\}$, whose Haar measure is a fair coin. Independent flips preserve diagonal coordinate masses and randomise off-diagonal cross-coordinate signs, which cancel in expectation, while leaving $\Delta W \Delta W^\top$ and hence the spectrum untouched. On ViT-B/16 the singletons carry 71% of the stored eigenvalue mass, so skipping them retained substantial coupling.

The LoRA generator matched a norm where a factor spectrum was required. The factor-conditioned null of §3.3 preserves A and the singular values of B ; it does not generally preserve the product spectrum of BA . Its first implementation redrew B as an i.i.d. Gaussian at matched Frobenius norm, leaving even the factor spectrum uncontrolled: a Gaussian is nearly Marchenko–Pastur where trained B decays. On fitted-reference pools with shared A and independent B , the flat-spectrum generator reports $+0.33$ when B has no decay and -0.16 when it does. Drawing B at its observed singular values holds the reference offset within $+0.008$ across the same sweep. This removes the observed factor-spectrum mismatch; it is not a claim of exact product-spectrum preservation. Indeed the nonzero squared singular values of the redraw are the eigenvalues of $\text{diag}(S_B)W^\top(AA^\top)W\text{diag}(S_B)$, which depend on W unless $AA^\top$ is isotropic. A uniform product-spectrum-preserving rotation inside $\text{row}(A)$ fails differently: it discards this anisotropy and has constructed-reference offset $+0.2023$. Rotating only to BQ retains a learned singular frame and changes the product singular values by 19.7%.

J PROJECTOR REBUILD

[Qiu et al. \(2026\)](#) filter an incoming task vector with $P = I - VV^\top$, where V contains the top- r_p right singular vectors of the cumulative update τ . We build this basis once from the observations and once after independently applying the regime-specific null to every task. The activation null preserves full-update spectra and occupancies; the LoRA null preserves its factor-level invariants. The second column reports overlap of the resulting bases, while the fourth is a separate concentration statistic. Neither identifies how that structure arose, tests accuracy, or contradicts the gains reported for the projector. A basis reproducible under a conditional null cannot, without further evidence, identify the

Table 5: Projector-basis overlap after independent draws from each fitted conditional null, and the observed basis’s mass in the top-64 activation subspace. ViT-B/16, eight tasks, $r_p = 128$ over 12 full-fine-tuning modules and $r_p = 16$ over 24 adapter modules.

pool	observed-null	chance	in activation	chance
full fine-tuning	0.2594	0.1667	0.2612	0.0833
LoRA-16	0.8750	0.0208	0.1356	0.0833

cross-task mechanism used to justify it. The LoRA value has no projector-level positive control; the factor-null overlap stress test covers only one different, narrow intervention.

K WHICH CORRECTION DID THE WORK

The two corrections to the activation null can be switched independently. Scoring the same planted pairs at $s = 2$ with each combination separates them.

	full complement	singleton flip	level-centred recovery
			49.0%
		✓	77.7%
✓			89.2%
✓		✓	99.3%

Each arm is centred by its own $s = 0$ level and divided by the same delivered increment as in Eq. 6. The singleton flip alone raises recovery from 49.0% to 77.7%, the full complement alone to 89.2%, and both to 99.3%; the effects overlap. The singleton flip is not a token gesture. On ViT-B/16 the singletons hold 71% of the stored eigenvalue mass, and leaving them fixed exempted most of $\text{span}(V)$.

L PLANTED-STRUCTURE RECOVERY, BY LEVEL

Planting operator. Let $P_0 = U_0 \text{diag}(\sigma_1, \dots, \sigma_\rho) V_0^\top$ be a thin SVD with descending singular values, and let $S = V_A[:, 1:s]$ contain the source task’s first s right singular vectors. We set the first s columns of $\hat{V}$ to S . In order, we project $V_0[:, s+1:\rho]$ onto $S^\perp$ and apply QR/Gram–Schmidt against all columns already accepted; any algebraic or numerical rank deficiency is filled by a Haar frame in the remaining complement. We then set $P_s = U_0 \text{diag}(\sigma_1, \dots, \sigma_\rho) \hat{V}^\top$. Hence $\hat{V}^\top \hat{V} = I$ and every singular value and left singular vector of P_0 is retained. Away from a tie at the relevant singular-value boundary, the first s leading right directions match the source exactly. The same planted matrices are scored by every candidate null.

Table 6: Positive-control response on ViT-B/16, four modules, six tasks. *initial* is the first activation null and *corrected* is §3.2. Both score the same planted pairs. *delivered* is $(\text{raw}(s) - \text{raw}(0))/[(s/k)(1 - \mathcal{O}_0)]$. Recovery is the level-centred excess increment divided by the delivered overlap increment, Eq. 6; it is therefore comparable across planting levels and is not the uncentred excess divided by s/k .

planted	raw	delivered	initial recovery	corrected recovery
0/8	0.02771	—	$X_0 = +0.00004$	$X_0 = -0.000623$
1/8	0.14129	93.45%	45.6%	99.6%
2/8	0.26235	96.53%	49.5%	99.4%
4/8	0.50557	98.30%	51.7%	99.3%

Table 7: Cross-task read-subspace overlap and its scalar decomposition. Complete encoder pools use all 190 task pairs; the matched eight-task row uses all 28 pairs. `raw` is Eq. 2, `iso` is k/d , and `null` is the exact expectation in Eq. 4. Terms are averaged before computing the reproduced fraction. No Monte Carlo draw enters these scalar results. ViT-L/14 is run on `q` and `v` only.

	modules	raw	iso	null	residual	reproduced
ViT-B/16	72	0.104574	0.009115	0.033309	0.071265	25.3451%
ViT-B/32	72	0.095313	0.009115	0.028420	0.066893	22.3961%
ViT-L/14 <i>q/v</i>	48	0.071402	0.007813	0.015377	0.056024	11.8961%
matched FT8 <i>q/v</i>	24	0.089930	0.010417	0.023333	0.066597	16.2440%

M OVERLAP DECOMPOSITION, IN FULL

For the exploratory ViT-B/16 role grouping in Figure 2, the same unadjusted ratio is 16.5568% for residual-reading modules (`q`, `k`, `v`, `fc1`) and 38.9609% for block-internal readers (`out_proj`, `fc2`).

On the matched FT8 pool, a sampled-matrix validation gives mean null 0.02330, versus the exact 0.02333. Sampled matrices remain necessary for projector and positive-control experiments.

Table 8: External-reference H_0 stress diagnostics. These activation-generated references are not the exact Haar self-orbit, whose population $H_0 = 0$ by identity. The finite-draw offsets can mix reference mismatch and over-destruction and are never added to an estimate.

pool and candidate action	realised external H_0
matched FT8, activation null	−0.00040
LoRA-16, activation null	−0.00037
lin-LoRA, activation null	−0.00037

N THE GENERATIVE NULL AND WHY IT FAILS

The obvious construction is generative: draw a random update whose input-side covariance is proportional to the base activation covariance, adding an isotropic draw in the unresolved complement. It does not work because it must invent the relationship between the trained update and C_H . Independent draws then concentrate on the same high-variance activation directions even when observed updates have a different occupancy profile. On the explicitly scoped eight-task, eight-module common-reference diagnostic in Table 1, this complement-aware generator reports native raw-minus-null excess −0.8959, whereas the observed-profile-preserving null reports −0.0007. These are baseline-specific stress levels, not estimates of the headline conditional fraction. An earlier exploratory generator omitted the unresolved complement entirely; we discard its quantitative output rather than using a known-defective construction. The comparison locates the requirement precisely: a conditional null here must retain the observed update–activation occupancy rather than generate that relationship from an unverified model.